Unit
Plate Tectonics

Title
4. Convection in a Pan

Summary
What drives the motion of the Earth's tectonic plates? Partly, it is convection, the process by which heat energy is transferred by currents in a liquid or gas. Convection currents within the mantle carry tectonic plates along with the slowly moving mantle like giant rafts carried along by a current in a river. To help students understand this idea, soapy water in a pie pan is heated from below and convections currents can be observed forming and moving in the soapy water. Several prelude demonstrations help students recognize that hot things rise and cold things sink.

Objectives
Can recognize that hot things rise and cold things sink.
Can describe and explain the process of convection.
Can investigate and illustrate convection currents in a pan of soapy water.
Can correlate convection currents in the pan of water to convection currents in the Earth's mantle.
Can diagram convection currents in the Earth's mantle.

Vocabulary
Mantle
Lithosphere
Crust
Convection
Heat

Time
10 min demo and introduction
30-40 min investigation
10-15 min discussion

Grouping
Groups of 4-6 students

Materials
For the demo, the teacher needs:

- One dry cleaner bag
- Cellophane tape
- 3-4 paper clips
- Blow dryer (hand held hair dryer)

Each group of students needs:

- One aluminum pie tin
- Tea light candle, hot plate, or other heat source
- Matches or a lighter

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- 4 film canisters
- Food coloring in dropper bottles or with eyedroppers
- 2-3 cups water
- 2-3 tablespoons liquid hand soap or shampoo with a metallic, pearly appearance. SoftSoap® and Walgreen's Liquid Soap both work well. Look for glycol stearate, glycol distearate or glycerol stearate among the ingredients on the label.
- Small pocket-sized mirror

Every student needs a copy of the Convection in a Pan handout.

Setting
Classroom

Teacher Background
The earth has several major layers – a hot metallic core, a less hot liquid mantle, and the solid lithosphere and crust on top (see background section of Journey Through Earth for more information). The hot metallic core causes the mantle immediately above to heat up. As the liquid rock in the mantle heats up, it rises because a heated liquid (or gas) expands and becomes less dense than the cooler liquid (or gas) nearby. When this hot liquid reaches the top of the mantle layer, it gets pushed aside by more hot mantle rising below it, spreading out under the solid lithosphere above like a cloud of steam hitting the ceiling of the kitchen. As it spreads out, it cools. Cool liquids (and gases) shrink in volume and are more dense than the warmer liquids (or gases) nearby. Therefore, the cooled mantle sinks to the bottom of the mantle layer where it gets heated by the core and begins the cycle anew.

This process through which heat energy is transferred through currents within a liquid or gas is called convection. The cyclical nature of the process in an enclosed system like the mantle of the Earth results in convection cells – local regions of liquid or gas that form a relatively stable cycle (heating, rising, moving aside, cooling, and sinking in roughly the same location over and over again).

Convection takes place in many other systems. A pot of water boiling on the stove is a good example of convection. Watch spaghetti boiling in a large pot and you will see the noodles rise near the middle of the pot above the flames, spread out over the surface, and fall again near the edges where it is cool. In the Earth's atmosphere, convection results in regional weather patterns and thermals (rising columns of heated air). Eagles and hang gliders both take advantage of thermals to stay aloft. In the Earth's oceans, the warm ocean water near the equator tends to follow currents towards the poles while cold polar ocean water follow currents back again to the equator.

A key concept is that hot fluid and gas rises and cold fluid and gas sinks. To demonstrate this principle, you can create a hot air balloon in the classroom using a dry cleaner bag and a hand-held hair dryer.

Student Prerequisites

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Students should have learned about the Earth's layers. Ideally, they will also have learned about density and the relationship between temperature, volume, and molecular motion. However, this lesson is written assuming that students don't know about density and heat yet.

Getting Ready

Test the hot air balloon demo

1. Use cellophane tape to seal the top seam and any holes in the dry cleaner bag. Use as little tape as possible.
2. Clip 3-4 paper clips around the bottom edge of the bag, as evenly distributed as possible. This will keep the bag upright and stable as it takes flight.
3. Hold the bag at the top. Get helpers or use 2 chairs to keep the bottom edges open.
4. Turn the blow dryer on at the lowest setting and hold it near the bottom of the bag (but not so close that the bag begins to melt). Allow the hot air to inflate the bag.
5. When the bag is fully inflated, let go of it to test its buoyancy. If it lifts off, let it go. If it doesn't continue filling with the blow dryer for a little longer.
6. Watch its flight, taking note of how stable it is in the air. When it lands again, adjust the paper clips to make it more stable. If it tilts one way, move them to different locations. If it flips over completely, add more paper clips.

Set up the pie pans

1. Make copies of the Convection in a Pan student handout.
2. Each pie pan will need about 2-3 cups of water. In a large pitcher or bucket, fill the bucket with as much water as you need for all the groups (plus a few cups extra).
3. The dilution of soap to water is approximately 1 tablespoon soap to 1 cup water. Add the appropriate amount of soap to the water.
4. Mix the soap into the water gently, trying to minimize the number of bubbles. The final solution should be very fluid and should leave swirly trails as you move your spoon or hand through the solution.
5. Set the materials for each group into a pie pan: candle, 4 film canisters, matches, food coloring droppers.

Lesson Plan

Hot air balloon demo

1. Start class by asking for 4 volunteers.
2. Have 1 volunteer hold the top of the pre-tested bag with paper clips attached and 2 others to hold the bottom edges open.
3. Have the third volunteer turn on the blow dryer to the lowest setting and hold it under the opening. Make sure the blow dryer isn't too close to the opening that it melts the bag or overheats your volunteers hands.
4. As the bag fills, have the other students predict what will happen.
5. When the bag is full, have the blow dryer volunteer turn the dryer off.
6. On the count of three, have the other 3 volunteers let go and watch the balloon fly.
7. When the bag finally comes back down, discuss what happened. Some questions to consider include:

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- Why did the bag fly? What powered it?
- How is the hot air balloon the same or different than a helium balloon? Was helium used?
- What would happen if we used a regular fan blowing room temperature air into the bag? Why?
- What would happen if we let an air conditioning vent fill the bag? Why?
- Where is it hottest above a fire – directly above the flames or an equal distance to the sides of the flame? Why?
- What happens to the steam above a pot of boiling water? Where does it go? Why?
- If hot air rises, what do you think will happen to cold air?
- Do you think this happens in a liquid? Do you think hot water will rise among cooler water? How about in a solid?

Convection in a pan exploration

1. Tell students that they are going to watch what happens when a candle is placed under a pan of soapy water. Get the students into groups and assign each a work area to assemble around.
2. Have one member of each group get a pie pan and the other needed materials.
3. For the candle version of the activity, spread out the 4 film canisters on the table. Place the pie pan on top. It should look like a circular table on peg legs.
4. Go around to each group and fill each pan a little more than half full with soapy water. Tell the students what is in the water to make it pearly. Ask them not to touch the surface so that the liquid can settle and fluid motion can cease.
5. Pass out the handouts. Tell students that they are going to light the candle and put it under the pan. Something will happen to the liquid. Explain that their job is to carefully watch the liquid and draw arrows on the diagrams in the handout to show how the liquid is moving in different places.
6. When the water in the pans are still, have students light the candle and slip it underneath the pan, right in the center.
7. Have students watch what happens. The food coloring is a tool that can help them figure out what is going on by placing a drop in different places around the pan and watching how the coloring moves away from that spot.
8. Let the students experiment, discuss and draw for up to 10 minutes then blow out the candles. After 10 minutes or so, the water in the pan will begin to get too hot and the pearlescent molecules (glycol stearate) will begin to break down.
9. Give students a little more time to finish their drawings.
10. Optional: See the variations suggested in the going further section.
11. Clean up and dispose of the soap solution down the sink (or save it for the next class).
12. Discuss the students' observations of convection currents together as a class. Together, draw 3 different views of the fluid motion on the board (top view, bottom view, and side view.
13. Relate this activity to what the students saw with the hot air balloon. Hot air rises. So how does that relate

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TIFF (LZW) decompressor
are needed to see this picture.

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to what is going on in the pie pan? My students found it helpful to label the side view drawing of the pie pan convection cells with labels describing what is happening at every place. (For example, label the water right above the candle in the center “Water near candle flame gets hot.” Label an arrow rising from the candle to the surface “Hot water rises.” Etc.)

14. Discuss what is going on in the mantle of the Earth. On the board, draw a cross section through the Earth like the diagram shown. Relate this picture to the convection cells observed in the pie pan. Show how the super hot core of the Earth is like the candle heating the mantle above it until the hot mantle rises towards the surface of the Earth.
15. Optional: Discuss other examples of convection in the Earth’s atmosphere and oceans.

Assessment

1. Collect the students’ handouts and drawings.
2. In teams, have students predict what would happen in the following scenarios.
    a. You put a pot of cool water on the stove. You turn on the flame below the pot. The water at the bottom of the pot nearest the flame begins to get hot. Will convection currents be created? If so, diagram them, showing how the convection currents would move through the water. If not, explain why not.
    b. You have a pot of cool water sitting on the table. You turn on a heat lamp above the pot. The water at the top of the pot nearest the heat lamp begins to get hot. Will convection currents be created? If so, diagram them, showing how the convection currents would move through the water. If not, explain why not.
    c. You close all the windows and doors of the classroom. You set a portable camping stove on the floor in the middle of the room and turn the burner on. The air near the burner gets hot. Will convection currents be created? If so, diagram them, showing how the convection currents would move through the air in the room. If not, explain why not.
    d. You have a rectangular pan full of water. You balance the pan on a pedestal. Under one end of the pan you light a candle. The water near that end gets hot. Under the other end of the pan you place a large pot of ice water. The water near that end gets cold. Will convection currents be created? If so, diagram them, showing how the convection currents would move through the water. If not, explain why not.
    e. Think about the Pacific Ocean. The water near the equator is warmed by the sun. The water near the North and South Poles gets less direct sunlight and is very very cold. Will convection currents be created? If so, diagram them, showing how the convection currents would move through the water. If not, explain why not. (Hint, look at your answer to the previous question.)

Going Further

1. Try the following variations:
    - Try the activity with multiple candles.
    - Sprinkle baby powder on the surface and watch the particles move.

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- Use tinfoil to shape a small, flat foil raft to represent tectonic plates. Place one or more rafts on the surface of the pan with the candle lit to model the movement of tectonic plates. (Aluminum foil works better than Styrofoam and plastic since multiple rafts will not stick to one another through static forces, and if shaped with an out-turned lip, will not stick to one another through surface tension either).
- Once the solution is hot (has been used for 10 minutes), blow out the candle and put a piece of ice in the middle of the liquid.
- Once the solution is hot, very carefully pick the whole pan up and set it down on a cool table.
- Once the solution is hot, carefully slide a fifth film canister full to the brim with ice water under the pan. The surface of the ice water should contact the bottom of the pan.
- Try the activity with a candle at one end next to a fifth film canister full to the brim with ice water a little distance away.

2. This lesson does not discuss the relationship between temperature and volume or the concept of density. The only concept students are expected to understand is that hot things rise and that cold things sink. So study into these other concepts.
3. Try the variation of this activity used in a UC Museum of Paleontology teacher workshop (http://www.ucmp.berkeley.edu/education/dynamic/session1/sess1_act2.html). They use a rectangular baking dish full of water with a heat source at one end and a bucket of ice water at the other.

Sources

The activity elaborates upon one of the Exploratorium's science snacks "Pie Pan Convection" (http://www.exploratorium.edu/snacks/pie_pan_convection/index.html).

The hot air balloon demonstration was taken from a lesson by Gregory Vogt, edited by Roger Storm of the NASA Glenn Research Center (http://www.grc.nasa.gov/WWW/K-12/TRC/Aeronautics/Hot_Air_Balloon.html). See their lesson plan for detailed information, diagrams and tips.

A great resource for learning more about heat, density and convection is the first session of a UC Museum of Paleontology teacher workshop (http://www.ucmp.berkeley.edu/education/dynamic/session1/index.html). There are excellent, detailed descriptions of heat, density and convection concepts as well as teaching tips and resources.

Standards

Grade 6

Plate Tectonics and Earth's Structure

1. Plate tectonics accounts for important features of Earth's surface and major geologic events. As a basis for understanding this concept:

b. Students know Earth is composed of several layers: a cold, brittle lithosphere; a hot, convecting mantle; and a dense, metallic core.

c. Students know lithospheric plates the size of continents and oceans move at rates of centimeters per year in response to movements in the mantle.

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Heat (Thermal Energy) (Physical Sciences)
3. Heat moves in a predictable flow from warmer objects to cooler objects until all the objects are at the same temperature. As a basis for understanding this concept:
a. Students know energy can be carried from one place to another by heat flow or by waves, including water, light and sound waves, or by moving objects.

Energy in the Earth System
4. Many phenomena on Earth's surface are affected by the transfer of energy through radiation and convection currents. As a basis for understanding this concept:
c. Students know heat from Earth's interior reaches the surface primarily through convection.
d. Students know convection currents distribute heat in the atmosphere and oceans.

Grades 9-12 Earth Science
Energy in the Earth System
5. Heating of Earth's surface and atmosphere by the sun drives convection within the atmosphere and oceans, producing winds and ocean currents.

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